CODESCHOOL · PROPOSAL FOR [SCHOOL NAME] · DRAFT

Online Programme

Computational Thinking, coding and AI for Classes 3-8 on shared school devices, with automatic checking, a teacher dashboard and report-card export.

Prepared for: [Principal name], [School name], [City]

Prepared by Anoop Dobhal, CodeSchool · [Phone] · [Email]

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Aligned to CBSE's CT&AI curriculum for Classes 3-8 (2026-27)

Contents

Draft

This is a draft for discussion with [School name]. Scope, timeline, technical requirements and prices are indicative and will be finalised after the pilot discussion. Nothing in this document is a commitment until both sides sign an agreement, including a data processing agreement. A separate Offline Programme proposal is available for schools without student devices.

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Covering letter

[Date]

Dear [Principal name],

Thank you for considering CodeSchool. From the 2026-27 session, CBSE asks every affiliated school to teach Computational Thinking and Artificial Intelligence (CT&AI) in Classes 3-8: 50 hours a year in Classes 3-5 and 100 hours a year in Classes 6-8, including 40 hours of projects. CBSE's free handbooks give your teachers a good set of puzzles, but they leave out coding, project materials, hands-on AI before Class 8, and any way to track and report progress.

This proposal describes our **Online Programme**. It uses the school's existing computers or tablets, shared by pairs of students, to add what paper cannot: instant checking and hints, block coding on screen, AI models that students train and test themselves, and a live class dashboard with a report-card export. Every lesson also has a projector and paper fallback, so a lesson never stops because of a device or network problem.

We propose to begin with a **free pilot** in two or three sections from January to March 2027, with clear success measures agreed in advance and a data processing agreement signed before any student account is created. If the pilot meets your expectations, the programme can run in all the classes you choose from April 2027.

We would welcome your feedback on this draft and would be glad to meet you, your coordinator and your IT contact to tailor it to [School name].

Warm regards,

Anoop Dobhal, CodeSchool · [Phone] · [Email]

1. Proposal at a glance

Item	Proposal
Programme	CodeSchool Online Programme for Computational Thinking, coding and AI
Classes	3-8
What students use	An Android app and browser platform with level practice, block coding, an AI lab (Classes 6-8) and project workspaces
What teachers get	Teacher mode for the projector, a class dashboard, period-by-period scripts, a progress journal, report-card export, orientation and support
Devices needed	Shared school devices: about 1 per 2 students during the lesson, plus a projector for teacher mode
Internet	Needed to download lessons and sync results; lessons keep working offline
Time	Within CBSE's existing allocation: about 1-2 periods a week in Classes 3-5 and 3-4 periods a week in Classes 6-8
Pilot	Free, 8-10 weeks, January to March 2027, in 2-3 sections (for example one Class 5 and one Class 7 section)
Full programme	From April 2027 (session 2027-28), in the classes the school chooses
Indicative price after the pilot	Rs 299 + GST per student per year (school licence), or online access with the Rs 449 / Rs 599 book-list packs
Student data	School-managed accounts using roll numbers, no names required; CodeSchool acts only on the school's instructions

2. About CodeSchool

CodeSchool is an Indian education start-up building a level-based programme that helps schools teach computational thinking, coding and AI in Classes 3-8. The programme is being designed specifically for CBSE's 2026-27 CT&AI curriculum and for real Indian classrooms, where devices are usually shared, internet can be unreliable and teachers have little preparation time.

Our design is based on:

- a detailed review of **all twelve CBSE CT&AI student and teacher handbooks** for Classes 3-8 and CBSE's April 2026 webinar;
- international computer-science standards, including the **2026 CSTA PK-12 standards** and the **AI4K12 "Five Big Ideas in AI"**, which show the order in which children learn these ideas best;
- research on teaching programming, including the **PRIMM** lesson structure (predict, run, investigate, modify, make), **Parsons puzzles** and block-based coding;
- India's **Digital Personal Data Protection Act, 2023** and its Rules, which shape how we handle children's data from the start; and
- conversations with principals, coordinators and teachers at partner schools, which continue through the pilot.

3. Understanding the school's needs

What CBSE asks schools to do from 2026-27

	Classes 3-5	Classes 6-8
Hours per year	50, inside Mathematics and The World Around Us (TWAU)	100: 40 CT, 20 AI, 40 projects (two projects of 20 hours)
Who teaches	Maths and other subject teachers	Subject teachers for CT; the computer teacher for AI and projects
What CBSE provides free	Puzzle handbooks that follow the NCERT maths chapters, with one or two unplugged activities per class	The same, plus four AI chapters per class (concepts in the student book, answer keys in the teacher book)
Assessment CBSE expects	Written tests with CT puzzles, group activities, a Teacher Observation Journal	Written tests, practical examinations, group activities, projects, a Teacher Observation Journal

What teachers still need

Gap in the free handbooks	What the Online Programme adds
No coding: students follow algorithms but never write one	Block coding levels for every chapter, building to simple Python in Class 8
Hard puzzles with no feedback	Instant checking, hints after repeated mistakes and challenge levels
No project briefs, rubrics or examples for the 40 project hours	Two complete project kits per class with datasets, workspaces and rubrics
AI teacher books contain answer keys, not lesson scripts	A teacher script and a browser activity for every AI chapter
Hands-on AI only in Class 8, and only with webcams and internet	An AI lab from Class 6 using supplied datasets, with no webcam needed
A Teacher Observation Journal with no template or report-card format	A live dashboard, a competency journal and a report-card export

> **Our approach:** we do not replace CBSE's free books or the NCERT maths textbook. We sit on top of them, chapter by chapter, and add the missing pieces, so teachers use one joined-up plan.

4. Objectives and expected outcomes

Objectives

1. Let every student practise CT and coding at their own level, with instant feedback and hints.
2. Move students from **following** algorithms to **building** working programs, starting in Class 3.
3. Give Classes 6-8 hands-on AI every year: train a model, test it and find its bias.
4. Give Classes 6-8 complete, assessable project kits for their 40 project hours.
5. Give teachers and the coordinator a live view of progress and a report-card-ready summary.

What students will be able to do by the end of the year

Classes	Expected outcomes
3	Build short block programs with sequences, repeat loops and events; debug a program that does not work
4	Use if-then-else choices and nested loops; predict what a program will do before running it
5	Use variables and inputs; compare two ways of solving a problem; build a simple game with a score
6	Create custom blocks (functions); work with lists and charts; train and test a simple classifier; complete two projects
7	Use data to make predictions; build and compare classification and clustering models; rebalance a biased dataset; complete two projects
8	Read, change and run simple Python alongside blocks; follow the AI project cycle with a no-code model; audit a model for fairness; complete two projects

5. Scope and deliverables

Included in the programme

Component	Description	Classes
Levels app	Short levels for each NCERT maths chapter with automatic checking, hints and challenge levels	3-8
Block coding	"Now you build it" coding levels that turn each chapter's puzzle into a program	3-8
Code view	Students see and change the Python behind their blocks	7-8
AI lab	Browser activities where students train and test simple models using supplied datasets; no webcam required	6-8
Project kits	Two 20-hour projects a year: brief, datasets, online workspace, worked examples and rubrics	6-8
Teacher mode	Run any lesson on one projector for whole-class teaching	3-8
Class dashboard	Who has finished which level, who is stuck, and class-wide trends	3-8
Progress journal and export	Competency records mapped to CBSE CT&AI, exportable for report cards	3-8
Teacher scripts, orientation and support	Period plans, an at-school orientation, workshops and weekly support during the pilot	All
Paper fallback	Printable versions of every lesson for days without devices	3-8

Not included

- Hardware, devices, internet connections or lab set-up.
- A generative-AI chat tool for students.
- Replacing CBSE's handbooks or the NCERT maths textbook.
- Board-examination preparation for Classes 9-12.
- Any parent-paid home plan during the pilot.

6. Programme design

Levels that follow the maths chapters

Every lesson is tied to the NCERT maths chapter the class is studying. Students move through short levels: a **warm-up**, a **guided level**, a **build level** and a **challenge**. The platform checks answers instantly, offers a hint after repeated mistakes, and lets fast students move ahead. A short placement check at the start of the year puts each student at a suitable level, while the teacher can keep the whole class on the same chapter. The full chapter-by-chapter map is in **Appendix A**.

One routine for every lesson (about 40 minutes)

Step	What happens	Minutes
1. Predict and run	Students predict what a short program will do, then run it	8
2. Investigate	Tracing questions, checked automatically	8
3. Modify	Students change the program to do something new	8
4. Parsons puzzle	Drag jumbled blocks into the right order	8
5. Make	Pairs build their own program and share it with another pair	6-8

Steps 1-4 are checked automatically; step 5 is creative work the teacher can view on the dashboard. A complete sample lesson (shown in its paper form, which is also the fallback) is in **Appendix E**.

Pair work on shared devices

Lessons are designed for two students per device. Partners swap the "driver" and "navigator" roles at each step, so both students type, think and explain. Each student signs in with their own roll-number account at the start of the period on shared devices, or the pair's work is credited to both partners, as the school prefers.

Every child can take part

- **Different paces:** placement checks, hints and challenge levels mean no student is left waiting or left behind.
- **Language:** the app interface and teacher scripts can be provided in Hindi as well as English (to be confirmed for the school's medium of instruction).
- **Accessibility:** large touch targets, read-aloud instructions for younger classes, and colour choices that work for students with colour-vision differences.
- **Inclusion:** the teacher can assign easier or extension levels to individual students from the dashboard.

7. Class-by-class overview

Class	Computational thinking and coding	AI	Projects
3	Sequences, loops and events in block coding; cipher and path puzzles from the maths chapters	Sorting by features (enrichment)	Short in-app challenges
4	If-then-else choices; nested loops; debugging	Spotting patterns in examples (enrichment)	Short in-app challenges
5	Variables and inputs; search strategies; a capstone game	Training a simple picture sorter (enrichment)	Short in-app challenges
6	Custom blocks (functions); lists and data from the Data Handling chapter	What AI is; train and test a classifier with a supplied dataset; digital ethics	Weather across states; a smarter school routine
7	Data tables, charts and prediction; reading the code behind the blocks	Classification, clustering and bias: rebalance a dataset and retrain	Nutri-Quest food planner; sort it with AI
8	Reading and changing simple Python alongside blocks	The AI project cycle; no-code model building; guided versions of CBSE's Class 8 labs	Sports participation data; fair AI audit

Detailed maps: chapters in **Appendix A**, AI chapters in **Appendix B**, projects in **Appendix C**.

8. Annual plan (session 2027-28)

The programme fits inside CBSE's allocation. A typical school year of about 32 teaching weeks is planned as follows; the school can move units between terms.

	Term 1 (April-September)	Term 2 (October-March)	Total
Classes 3-5: CT and coding within Maths/TWAU	About 26 hours: placement check and first half of the maths chapters	About 24 hours: second half of the chapters plus the capstone game	50 hours
Classes 6-8: CT and coding	About 20 hours	About 20 hours	40 hours
Classes 6-8: AI lab	About 10 hours (AI chapters 1-2)	About 10 hours (AI chapters 3-4)	20 hours
Classes 6-8: Projects	Project 1: about 20 hours	Project 2: about 20 hours	40 hours

9. Sample weekly timetable

Class	Periods per week (40 minutes)	Example
3-5	1-2, taken from Maths or TWAU periods	Tuesday: teacher mode on the projector to introduce the level; Friday: pairs practise on tablets or in the lab
6-8	3-4	Monday: CT levels in the lab (maths teacher); Wednesday: AI lab (computer teacher); Thursday and Friday: project work (computer teacher)

If lab time is limited, the class can alternate: one lab period for practice and one classroom period with teacher mode and paper levels.

10. How the online programme runs

Element	Classes 3-5	Classes 6-8
Where	Computer lab, tablets in class, or projector with students taking turns	Computer lab or tablets; projector for AI demonstrations
Who teaches	Maths or EVS teacher	Subject teachers for CT; computer teacher for AI and projects
Devices	About 1 device per 2 students during the lesson (pair work)	About 1 device per 2 students during the lesson
Network	Lessons download in advance and work offline; results sync when online	Same; AI lab activities need a browser and a connection while training
Preparation time	About 5-10 minutes per lesson using the script and dashboard prompts	About 10 minutes per lesson; projects have a week-by-week plan
Fallback	Printable lesson or projector teacher mode	Printable lesson or projector teacher mode

11. Technical onboarding and support

Technical requirements (to be confirmed in a device check)

Requirement	Minimum
Android tablets or phones	Android 9 or later, 2 GB RAM, about 500 MB free space
Computers	A current Chrome, Edge or Firefox browser on Windows, Linux or ChromeOS
Projector	Any projector or smart board with a connected teacher device
Internet	Any connection for downloads and syncing; continuous internet is not required
Website access	The CodeSchool app and website must not be blocked by the school's filter

Onboarding steps

Step	Who	Time
1. Device check: test a sample of devices, the network and the filter	IT contact with CodeSchool (on a call or visit)	About 30 minutes
2. Roster upload: roll number, class and section for pilot students	Pilot lead, using a CodeSchool spreadsheet template	About 20 minutes
3. App installation or browser shortcut on lab devices	IT contact, with a step-by-step guide	About 5 minutes per device
4. Teacher accounts and trial login	CodeSchool sets up; teachers test	About 10 minutes
5. First lesson rehearsal during the orientation	Teachers with CodeSchool	Part of the 2-hour orientation

Support targets during the pilot

Issue	How to report	Target response
A lesson cannot start (urgent)	Phone or WhatsApp to the CodeSchool pilot contact	Within 1 hour during school hours, with the paper fallback used meanwhile
A bug or a wrong answer key	WhatsApp or email	Acknowledged the same day; fixed within 3 working days
A question about teaching or the dashboard	Shared question channel	Answered within 1 working day
A request for a change	At the weekly check-in	Discussed and planned at the next check-in

12. Implementation plan

Phases

Phase	When	What happens	Output
1. Agree scope	December 2026	Meet principal, coordinator and IT contact; choose pilot sections; device check; sign the pilot letter and data processing agreement	Signed pilot letter and agreement
2. Set up	Early January 2027	Parent notice and consent; roster upload; app installation; 2-hour teacher orientation	Accounts ready; teachers trained
3. Pilot	January to mid-March 2027 (8-10 weeks)	Lessons run; CodeSchool monitors usage and checks in weekly; the dashboard shows progress	Lessons delivered; progress data
4. Review	Mid-March 2027	Compare results with the agreed success measures; teacher and student feedback	Pilot report for the school
5. Roll out	April 2027 onwards	Accounts for all chosen sections; orientation for all teachers; termly workshops	Programme running in all chosen classes

Pilot week by week

Week	Classes 3-5 pilot section	Classes 6-8 pilot section	CodeSchool support
0	Orientation; first login; placement check (15 minutes)	Orientation; first login; placement check	Orientation at school; device check
1-2	Block coding levels for the current maths chapter (sequences and loops)	CT levels; AI lab chapter 1; project launch (define the problem)	Weekly check-in; usage report
3-4	Conditionals and debugging levels	AI lab: train and test a classifier; project data collection	Mid-pilot workshop (week 4)
5-6	Next chapter's levels; challenge levels	CT levels; project analysis with charts	Weekly check-in; usage report
7-8	Variables or nested loops (by class)	AI lab: bias and rebalancing; project testing and improvement	Assessment workshop (week 8)
9	Mini capstone game	Project reflection	Weekly check-in
10	End check; report-card export trial	Project presentations; end check; export trial	Review meeting and pilot report

Book-list and budget timing: most schools finalise next year's book list and budget in December-January, before the pilot ends. Schools that want the programme from April 2027 can decide after a demonstration and a trial login in December; the pilot will still shape how it is used.

13. Teacher support and professional development

- **Orientation (2 hours, at school):** the lesson routine, the teacher dashboard, teacher mode on a projector, and what to do when devices or internet fail.
- **Scripts in every lesson:** timings, what to say, answers and common mistakes.
- **Dashboard prompts:** which students are stuck and which level the class should do next.
- **Weekly check-ins** (15 minutes, by phone or WhatsApp) during the pilot, plus a shared question channel.
- **Two pilot workshops** (60 minutes each) and **one workshop per term** in the full year.
- **Short how-to videos** inside the teacher view.
- **Certificate of completion** for teachers who complete the orientation and termly workshops.
- **No specialist needed:** maths and EVS teachers can run Classes 3-5; the computer teacher leads AI and projects in Classes 6-8, as CBSE suggests.

The full plan and orientation agenda are in **Appendix F**.

14. Assessment and reporting

- **Automatic level checks:** every level is scored instantly, with mastery shown per chapter.
- **Progress journal:** competency records using a four-level rubric (beginning, developing, secure, extending) for abstraction, pattern recognition, decomposition and algorithmic thinking, plus AI literacy in Classes 6-8, with space for teacher observations.
- **Project rubrics:** six criteria, including testing and fairness; teachers score on screen or on paper.
- **Report-card export:** a term summary per student, ready to copy into the school's report card.
- **Baseline and end checks:** a 15-minute puzzle check at the start and end of the pilot and the year to show progress.
- **Coordinator view:** a school-level summary across sections, with no ranking of individual students.

Rubrics and an example report-card summary are in **Appendix D**.

15. Parent engagement

- **Parent notice and consent form** in English and Hindi, supplied by CodeSchool and sent by the school, explaining what the programme does and what data is used.
- **Information session:** an optional 20-minute session at a parent-teacher meeting, with a live demonstration.
- **Showcase:** at the end of each term, students present their projects or games to parents.
- **No home login is required** and no parent-paid plan is offered during the pilot. Any future home access would be optional and agreed with the school first.

16. Child safety and data protection

CodeSchool is designed to work as a **data processor for the school**. The school decides what data is used and remains in control of it, in line with India's Digital Personal Data Protection Act, 2023 and its Rules, whose main obligations, including those for children's data, are expected to take full effect in May 2027.

Principle	What it means in practice
Minimum data	Accounts use the school's roll number, class and section. No photos, addresses, Aadhaar numbers or dates of birth
School in control	The school creates and removes student accounts; students cannot sign up on their own
Parent consent	Collected by the school using its admission records, with a notice and form supplied by CodeSchool in English and Hindi
No tracking or advertising	No advertising of any kind, no third-party trackers, and no behavioural profiling of students
Learning records only	We record scores on curriculum levels that teachers can see, not behaviour
No AI training on student work	Student work is never used to train AI models
No open chat or sharing	Students cannot message each other or anyone outside the class; shared work is visible only within the class
No camera or microphone needed	AI activities use supplied datasets, so no images or voices of students are collected
Data in India	Data will be stored in an Indian cloud region and protected by access controls and encryption
Deletion	Student data is deleted at the end of the agreement, or earlier on the school's request
Breach response	Prompt notice to the school, and reporting as Indian law requires
Written agreement	A data processing agreement is signed with the school before any student data is shared

17. Governance, monitoring and quality

Mechanism	Frequency	Who	Purpose
Pilot lead check-in	Weekly during the pilot	Coordinator and CodeSchool	Progress, issues, next steps
Usage and mastery report	Weekly during the pilot; monthly afterwards	CodeSchool to the coordinator and principal	Active students, levels completed, sections needing support
Review meeting	End of pilot and end of each term	Principal, coordinator, CodeSchool	Results against success measures; changes needed
Classroom visit	Once during the pilot, with permission	CodeSchool	Support teachers and improve the platform
Data protection review	Before the pilot and once a year	IT contact or data lead, and CodeSchool	Check accounts, access and deletion against the agreement
Escalation	As needed	Coordinator to CodeSchool founder	Any concern is answered within two working days

18. What we need from the school

Need	Detail
A pilot lead	The coordinator or a senior teacher as our single point of contact
An IT contact	Someone who can install the app, check website access and support the device check
Teachers	One named teacher per pilot section, available for the orientation and two workshops
Devices and timetable	Access to the lab or tablets for the agreed periods each week
Class rosters	Roll number, class and section for pilot students (names optional and kept by the school)
Parent consent	The school informs parents and records consent using its admission records; CodeSchool provides the notice and form
Feedback	Two short feedback conversations (mid-pilot and end)

19. Pilot success measures

These are agreed in advance so the pilot result is clear to both sides. Targets are suggestions and can be changed.

Area	Measure	Suggested target
Teacher delivery	Planned pilot periods the teacher ran without outside help	80%
Fit	Lessons that fit within one period	90%
Learning	Students reaching mastery on each chapter's levels	70%
Progress	Average improvement from baseline to end check	A clear gain, reported by class
Use	Pilot students active in the platform each week	80%
Reliability	Lessons completed despite device or network issues (using fallback if needed)	95%
Teacher view	Would use it again next session (1-5)	4 or more
Reporting	Report-card export usable by the coordinator	Yes

20. Investment

Pilot terms

- The pilot is **free** for 2-3 sections: platform access, orientation, workshops and weekly support.
- There is **no obligation** to buy afterwards.
- Pilot student data is deleted at the end of the pilot unless the school decides to continue.

After the pilot (indicative)

Option	What is included	Indicative price
School licence	Platform access for students, teacher mode, dashboard, journal and export, project kits, teacher scripts, orientation, termly workshops and support	Rs 299 per student per year + 18% GST, paid by the school
Book-list pack with online access	Printed workbook plus school-activated online access, project kits and journal	Rs 449 (Classes 3-5) or Rs 599 (Classes 6-8) per student per year (MRP), on the school's book list

How payment works

- **School licence:** invoiced to the school once a year, based on the number of students enrolled in the chosen sections at the start of the session; students who join mid-year are added at no extra charge.
- **Book-list pack:** the pack goes on the school's **published book list** and can **replace the existing Computers book**, so the total cost to parents stays about the same. Packs are also available from booksellers and online, so **no parent is required to buy from one source**; access is still activated through the school's accounts.
- **No new fee head** is needed in either option.
- Prices are indicative, will be confirmed after the pilot, and will be held for the 2027-28 session once confirmed.

21. Roles and responsibilities

Task	CodeSchool	School
Platform, lessons, kits and scripts	Provides and updates	Reviews and gives feedback
Curriculum mapping to the school's chapter order	Prepares	Confirms
Device check and app installation	Guides	IT contact installs
Student accounts	Sets up from the roster	Provides roster; manages changes
Parent notice and consent	Supplies notice and form	Informs parents and records consent
Teacher orientation, workshops and support	Leads	Releases teachers to attend
Teaching the lessons	Supports	Delivers
Data protection	Acts as processor under the agreement	Acts as data fiduciary; approves data use
Pilot review and report	Prepares	Reviews and decides

22. Risks and how we manage them

Risk	How we manage it
Devices are shared or some do not work	Pair work, teacher mode on a projector, and a printable version of every lesson
Internet is slow or fails	Lessons download in advance, work offline and sync later
Lab time is limited	Alternate lab and classroom periods, using teacher mode and paper levels
Teachers have little time to prepare	Every lesson has a short script and dashboard prompts
Students use the app but do not progress	Weekly usage and mastery reports to the pilot lead, with suggested next steps
Concerns about children's data	Processor model, minimum data, no advertising or tracking, written agreement, Indian hosting
A teacher leaves or is absent	Scripts and the dashboard let another teacher take over; CodeSchool runs a catch-up orientation
CBSE publishes its own project list or rubrics	Project kits will be realigned to CBSE's topics as soon as they are published

23. Online or offline: choosing the right fit

Question	Online Programme	Offline Programme
Student devices needed?	Yes, shared devices (about 1 per 2 students during the lesson)	No
Where students practise	App or browser, with automatic checking	Paper workbook and classroom activities
Coding	Block coding on screen, with Python in Classes 7-8	On paper, with cut-out blocks
AI	Hands-on model training with supplied datasets	Unplugged card labs; projector demonstration
Teacher reporting	Live class dashboard and report-card export	Printable journal
Student data	School-managed accounts using roll numbers	None

Many schools may prefer a **hybrid**: the Offline Programme in Classes 3-5 and the Online Programme in Classes 6-8, where the computer lab is available. A separate Offline Programme proposal is available.

24. Frequently asked questions

Does this replace CBSE's free handbooks? No. It works alongside them, chapter by chapter, and adds what they do not include.

How many devices do we need? About one device for every two students during the lesson. A class of 40 needs about 20 devices, which can be the computer lab or a shared tablet trolley.

What if the internet goes down? Lessons are downloaded in advance and keep working offline; results sync later. Every lesson also has a projector and paper version.

Who teaches it? Your own maths and EVS teachers in Classes 3-5, and your subject and computer teachers in Classes 6-8, as CBSE suggests. No specialist is needed.

Will it take extra periods? No. It uses the hours CBSE has already allocated to CT&AI.

Is it safe for children? Students sign in with school-created roll-number accounts, there is no chat, advertising or tracking, no camera or microphone is used, and a data processing agreement is signed before the pilot.

How is it assessed? Through automatic level checks, the progress journal, project rubrics and a term summary for the report card (Appendix D).

Can students use it at home? Not during the pilot. Any future home access would be optional and agreed with the school first.

What if CBSE changes the curriculum? The programme is built around skills, with CBSE mapping kept separate, so lessons can be updated centrally and quickly.

Can we stop after the pilot? Yes. There is no obligation to continue, and pilot data will be deleted.

25. Assumptions and terms

- This proposal assumes the school follows CBSE's CT&AI curriculum and the NCERT mathematics textbooks for Classes 3-8.
- Pilot dates assume a signed pilot letter, a data processing agreement and a successful device check by mid-December 2026.
- Technical requirements will be confirmed in the device check.
- Prices are indicative until confirmed in writing after the pilot.
- CodeSchool retains ownership of its platform and materials; the school may use them for its own students during the agreement.
- Formal terms will be set out in a separate agreement.
- "CodeSchool" is the programme's working name.

26. Glossary

Term	Meaning
CT&AI	Computational Thinking and Artificial Intelligence, CBSE's subject for Classes 3-8 from 2026-27
Computational thinking	Solving problems in steps a person or computer can follow: abstraction, pattern recognition, decomposition and algorithms
Algorithm	A precise set of steps to solve a problem
Block coding	Programming by snapping together visual blocks instead of typing
Loop	An instruction to repeat steps
Conditional	An if-then-else choice in a program
Variable	A named store for a value that can change, such as a score
Function	A named set of steps that can be reused
Parsons puzzle	Putting jumbled lines of code in the right order
PRIMM	A lesson routine: predict, run, investigate, modify, make
Classifier	An AI model that sorts examples into groups it has learned
Bias (in AI)	Unfair results caused by unbalanced or incomplete data
Data processor	An organisation that handles data only on another organisation's instructions (here, the school's)
Data fiduciary	The organisation that decides why and how personal data is used (here, the school)

27. Next steps

1. A short meeting with the principal, coordinator and IT contact to agree the pilot sections.
2. A device check (about 30 minutes) and a trial login for teachers.
3. The school returns the response form below, and both sides sign the pilot letter and data processing agreement.
4. Parent notice, roster upload, teacher orientation and the pilot start in January 2027.

School response form

Fill this in on screen or on paper and return it to CodeSchool. It is an expression of interest, not a contract.

School name	
Board and city	
Contact person and role	
Phone / email	
Classes and sections for the pilot	
Approximate number of students	
Preferred start date	
Programme of interest	-
Questions or changes you would like	

Appendix A. Chapter-by-chapter curriculum map

CBSE's CT&AI handbooks follow the NCERT mathematics textbook chapter by chapter, so CodeSchool does the same. For every chapter, the map below shows the main computational-thinking (CT) skill CBSE's handbook practises and the CodeSchool level or activity that adds a "now you build it" step. **This is the planned mapping for the pilot; it will be confirmed with the school's teachers and adjusted if the school follows a different chapter order.**

CodeSchool levels are grouped in bands. **Band B (Builders)** covers Classes 3-5: B1 sequences and loops, B2 events and patterns, B3 conditionals (if-then-else), B4 nested and repeat-until loops, B5 variables and inputs, B6 CT puzzle strategies, B7 capstone game. **Band C (Makers)** covers Classes 6-8: C1 functions, C2 lists and data, C3 AI literacy I, C4 the code behind the blocks, C5 reading code, C6 writing simple code, C7 AI literacy II, C8 projects.

Class 3 (14 chapters, about 50 hours a year inside Maths and TWAU)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. What's in a Name?	Patterns; step-by-step encoding (Caesar cipher activity)	B1 "Secret messages": write and follow encoding steps
2. Toy Joy	Abstraction: views of 3D objects	B1 "Build from instructions": block towers from sequence cards
3. Double Century	Number sequences	B1 "Count-along loops": the repeat card
4. Vacation with My Nani Maa	Decomposition of multi-step problems	B1 "Plan the trip": order and debug a sequence
5. Fun with Shapes	Shape patterns, folding	B2 "Shape patterns with repeat"
6. House of Hundreds I	Place-value logic	B2 "Robot counter": events ("when tapped, add 10")
7. Raksha Bandhan	Equal groups, data sufficiency	B2 "Repeat the group"
8. Fair Share	Halves and fair rules	B2 "Split it fairly": sorting by a rule
9. House of Hundreds II	Clue-based reasoning	B2 "Clue detectives": which clues are enough?
10. Fun at Class Party!	Ordering by measurement	B2 "Line up by rule" (sorting)
11. Filling and Lifting	Rule with a branch ("on a white block skip two, on a black block skip one"); weighing logic	B3 preview "Skip rules": first if-then cards
12. Give and Take	Step-by-step exchanges	B2 "Trace the exchange": keeping count (variables preview)
13. Time Goes On	Sequences in time	B2 "Clock loops"
14. The Surajkund Fair	Paths, grids, pathway tiles	B1-B2 "Robot routes": path tiles and grid programs

Class 4 (14 chapters, about 50 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. Shapes Around Us	Views, faces, edges	B1 review: "Build from instructions"
2. Hide and Seek	Paths and turns; a rule with an exception	B3 "Robo makes choices" (sample lesson, Appendix E)
3. Patterns Around Us	Parity and error detection (Coin Flip Magic activity)	B2 "Find the flipped coin": checking rules
4. Thousands Around Us	Ordering (floor Sorting Network activity)	B4 "Sorting machines": compare-and-swap steps
5. Sharing and Measuring	Fair sharing, fractions	B3 "If it is even, then…"
6. Measuring Length	Shortest paths on tiles	B4 "Shortest route": loops with turns
7. The Cleanest Village	Tracing a button-press sequence	B3 "Button machine": trace and predict
8. Weigh It, Pour It	Comparing quantities	B3 "Heavier or lighter?": if-else decisions
9. Equal Groups	Repeated groups	B4 "Loops inside loops"
10. Elephants, Tigers and Leopards	Data and clues	B2 "Clue tables"
11. Fun with Symmetry	Reflections and turns	B4 "Mirror drawings": repeat with turns
12. Ticking Clocks and Turning Calendar	Cycles and counting	B4 "Repeat until": day and time cycles
13. The Transport Museum	Classifying vehicles by features	B3 "Sort by rules": decision cards
14. Data Handling	Tables and tallies	B2 "Count and chart"

Class 5 (15 chapters, about 50 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. We the Travellers I	Multi-step journeys	B4 review: routes with loops and turns
2. Fractions	Part-whole patterns	B5 "Keep score": variables that change
3. Angles as Turns	Turns and directions	B4 "Turtle turns": draw shapes with angles
4. We the Travellers II	Planning with constraints	B5 "Fuel left?": inputs and variables
5. Far and Near	Measuring and comparing	B5 "Compare and decide"
6. The Dairy Farm	Large quantities, grouping	B5 "Counters and totals"
7. Shapes and Patterns	Growing patterns	B4 "Pattern generators"
8. Weight and Capacity	Searching with a weighing scale (linear vs divide-and-conquer)	B6 "Search strategies": which search is faster?
9. Coconut Farm	Multiplication patterns	B5 "Harvest calculator"
10. Symmetrical Designs	Reflection and repetition	B4 "Design machines"
11. Grandmother's Quilt	Tiling patterns	B6 "Tile puzzles": decomposition
12. Racing Seconds	Timing and ordering	B5 "Stopwatch": variables over time
13. Animal Jumps	Skip patterns on number lines	B6 "Jump rules": loops with conditions
14. Maps and Locations	Grids and directions	B7 capstone "Treasure map game" (design)
15. Data Through Pictures	Pictographs and tables	B7 capstone: add a score and finish the game

Class 6 (10 CT chapters plus 4 AI chapters; 100 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. Patterns in Mathematics	Number and shape patterns	C1 "Make a pattern block": functions
2. Lines and Angles	Angles, constructions	C1 "Draw with functions": reusable steps
3. Number Play	Number rules and puzzles	C1 "Number machines"
4. Data Handling and Presentation	Tables, charts	C2 "Lists and charts"
5. Prime Time	Factors; 12-point circle step-rule activity	C2 "Prime finder": loops over lists
6. Perimeter and Area	Measurement rules	C1 "Area calculator" with inputs
7. Fractions	Visual patterns of fractions	C2 "Fraction lists"
8. Playing with Constructions	Step-by-step constructions	C1 "Construction scripts"
9. Symmetry	Folding and transformation	C1 "Symmetry painter"
10. The Other Side of Zero	Negative numbers, number lines	C2 "Temperature log": data with negatives

Class 7 (8 CT chapters plus 4 AI chapters; 100 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. Large Numbers Around Us	Place value at scale	C2 "Big data lists"
2. Arithmetic Expressions	Order of operations	C4 "The code behind the blocks": expressions
3. A Peek Beyond the Point	Decimals; punched-card binary sorting activity	C4 "Binary and sorting"
4. Expressions using Letter-Numbers	Variables and algebra	C4-C5 "Variables in code"
5. Parallel and Intersecting Lines	Geometry rules	C5 "Read and predict": tracing programs
6. Number Play	Logic puzzle punch cards (true/false, if-then)	C5 "Logic and conditions"
7. A Tale of Three Intersecting Lines	Triangle rules	C5 "Debug the program"
8. Working with Fractions	Fraction operations	C4 "Fraction functions"

Class 8 (7 CT chapters plus 4 AI chapters; 100 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. A Square and a Cube	Powers and patterns	C5-C6 "Squares and cubes" in simple code
2. Power Play	Exponents	C6 "Repeat and multiply": loops in code
3. A Story of Numbers	Number systems; binary and ternary "pipes" activity	C6 "Number system converter"
4. Quadrilaterals	Classification by properties	C6 "Shape classifier" with if-else
5. Number Play	Logic and rules	C6 "Rule checker"
6. We Distribute Yet Things Multiply	Distribution and expressions	C6 "Expression tester"
7. Proportional Reasoning	Ratios and scaling	C6 "Recipe scaler" with variables

Appendix B. AI strand for Classes 6-8

CBSE's 20 AI hours a year in Classes 6-8 are organised into four chapters per class. CBSE's student books explain the concepts; the teacher books give exercises and answer keys. CodeSchool adds a teacher script and a hands-on activity for every chapter, in an offline form (cards, paper and one projector) and an online form (browser activities with supplied datasets). Schools can use either form.

Class 6

CBSE AI chapter	What students learn	Offline activity	Online activity
1. Introduction to AI and everyday examples	What AI is; AI vs automation; supervised, unsupervised and reinforcement learning	"Is it AI?" sorting cards; "human classifier" game with labelled pictures	Train a picture sorter with a supplied set of labelled images
2. Basic data concepts	Types of data; collecting and organising data	Class survey on paper; build a frequency table	Enter survey data and see charts update
3. Simple pattern recognition and decision making	Finding patterns; observations vs conclusions	Pattern cards and decision trees on chart paper	Build a decision tree and test it on new examples
4. Ethics and digital responsibility	Passwords, phishing, digital footprint, plagiarism	Scenario cards and class debate	Interactive scenarios with instant feedback

Class 7

CBSE AI chapter	What students learn	Offline activity	Online activity
1. AI domains and applications	Data science, computer vision, language; classification, regression, clustering	Card labs: classify fruits by features; group "unlabelled" cards into clusters	Train a classifier, a simple prediction line and a clustering tool on small datasets
2. AI in industries	Case studies of AI in health, transport and farming	Case-study cards and role-play	Explore case studies with short quizzes
3. Data visualisation and analysis	Bar, line and pie charts; reading trends	Hand-drawn charts from class data	Build charts from a dataset and spot trends
4. Ethics and AI bias awareness	How unbalanced data causes unfair results	"Lopsided deck" activity: a biased card set gives unfair answers	Retrain a model after rebalancing its data and compare results

Class 8

CBSE AI chapter	What students learn	Offline activity	Online activity
1. AI project lifecycle	Define the problem, collect data, test, reflect and improve	Plan an AI project on a lifecycle worksheet	Run the full lifecycle with a no-code model
2. AI and its applications	No-code AI tools; environment, health and education examples	Teacher-led demonstration of CBSE's waste-classifier lab on one projector	Guided version of CBSE's waste-classifier and voice-command labs, with supplied data
3. Data and fairness in AI	Bias in datasets; rebalancing	Card-based fairness audit	Guided version of CBSE's cat/dog bias lab: test, rebalance, retrain
4. Ethics and responsible AI	Privacy, misinformation, responsible use	Debate and "spot the fake" cards	Interactive misinformation and privacy scenarios

Appendix C. Project kits for Classes 6-8

CBSE allots **40 hours a year to two required projects of 20 hours each** in Classes 6-8, assessed by the computer teacher. CBSE has said a project list with rubrics is coming; until it is published, CodeSchool's kits follow the topics CBSE presented in its April 2026 webinar, and will be realigned to CBSE's official list as soon as it appears.

Every kit contains: a student brief, a teacher guide with a week-by-week plan, data sheets or datasets, a worked example, a marking rubric (Appendix D), and a presentation template.

Class	Project 1 (20 hours)	Project 2 (20 hours)
6	**Weather across states:** collect and compare weather data from different states, find patterns and present conclusions	**A smarter school routine:** design an algorithm and flowchart to solve a school problem (for example, library returns or lunch queues), then test it
7	**Nutri-Quest weekly food planner:** use nutrition data and rules to design a balanced week of meals	**Sort it with AI:** build a classifier for school waste (cards offline, a trained model online) and measure its accuracy
8	**Sports participation and performance:** collect sports data, analyse it and recommend changes	**Fair AI audit:** test a classifier for bias, rebalance the data and present recommendations

Standard 10-week project plan (about 2 hours a week)

Weeks	Stage	What students do	Evidence for assessment
1-2	Define	Understand the problem, form teams, plan questions	Problem statement and plan
3-4	Collect	Gather data or design the first version of the algorithm	Data sheet or first flowchart
5-6	Build	Organise data, make charts, build the algorithm or model	Charts, flowchart or model
7-8	Test and improve	Test with new cases, find errors, improve	Test log with changes made
9	Reflect	Discuss fairness, limits and what was learned	Short reflection
10	Present	Share findings with the class or at a showcase	Presentation and final product

Appendix D. Assessment rubrics and report-card format

CBSE asks for continuous, competency-based assessment for CT&AI, including written tests with CT puzzles, group activities, practical work, projects and a **Teacher Observation Journal**, but it has not published a template or report-card format. CodeSchool provides the following, mapped to CBSE's competency areas.

Competency rubric for the observation journal

Competency	Beginning	Developing	Secure	Extending
Abstraction	Needs help to ignore unimportant details	Picks out key details with prompts	Identifies the key details on their own	Explains which details matter and why
Pattern recognition	Spots simple repeating patterns with help	Continues patterns with one change	Finds and extends patterns with two or more changes	Creates and explains new patterns
Decomposition	Needs help to break a problem into steps	Breaks a familiar problem into parts	Breaks new problems into clear parts	Plans the parts and their order independently
Algorithmic thinking	Follows a short sequence with help	Follows and fixes simple sequences	Writes sequences with loops and conditions	Designs, tests and improves their own algorithms
AI literacy (Classes 6-8)	Names everyday uses of AI	Explains that AI learns from data	Trains or tests a model and explains its results	Identifies bias and suggests fairer data

Project rubric (used for every Class 6-8 project)

Criterion	1 Beginning	2 Developing	3 Secure	4 Extending
Problem definition	Problem unclear	Problem stated but vague	Clear problem and goal	Clear problem, goal and success measure
Data or algorithm	Little data or no clear steps	Some data or steps, with gaps	Complete data or working algorithm	Well-organised data or efficient, tested algorithm
Testing and improvement	Not tested	Tested once	Tested with new cases and improved	Systematic testing with recorded improvements
Ethics and fairness	Not considered	Mentioned	Discussed with an example	Changed the work to make it fairer
Communication	Hard to follow	Partly clear	Clear and organised	Clear, engaging and well supported by evidence
Teamwork	Little contribution	Some contribution	Shared work fairly	Led and supported others

Term summary for the report card (example)

Area	Grade	Teacher comment
Computational thinking (overall)	Secure	Breaks problems into steps and writes loops confidently
Coding (build levels completed)	12 of 15 levels	Ready for conditionals with variables
AI literacy (Classes 6-8)	Developing	Trained a classifier; working on explaining bias
Project	3 (Secure)	Clear charts; testing could be more systematic

Appendix E. Sample lesson: "Robo makes choices" (Class 4, Level B3)

This lesson pairs with the Class 4 NCERT chapter *Hide and Seek* and teaches if-then-else choices. It runs in one 40-minute period with no devices (a projector is optional). A full printable version with the student sheet is available on request.

Step	Minutes	Teacher does	Students do
1. Predict and run	8	Draws a row of five squares (START, PUDDLE, plain, PUDDLE, plain) and shows a program: repeat 4 times, move forward, *if* on a puddle *then* say "Splash!" *else* say "Step"	Write what Robo will say, then one student acts as Robo while the class reads the program aloud (answer: Splash, Step, Splash, Step)
2. Investigate	8	Asks four tracing questions, including "what if the if-else were outside the loop?"	Discuss in pairs and explain why the check happens every time
3. Modify	8	Draws a new row with STAR squares	Change only the if-else so Robo says "Star!" on star squares
4. Parsons puzzle	8	Hands out jumbled blocks, one of which is a trick	Put the blocks in order so Robo reaches the goal without hitting a wall
5. Make and exit ticket	8	Invites pairs to invent their own rule	Draw five squares, write "If Robo is on a ___ square, then ___, else ___", swap and predict; answer a two-question exit ticket

Common mistakes the script prepares teachers for: thinking both branches run; thinking the "if" is checked only once.

Appendix F. Teacher development plan

When	Session	Length	Content
Before the pilot	Orientation at school	2 hours	Why CT&AI; the five-step lesson routine; a live demonstration lesson; the observation journal; questions
Pilot weeks 1-10	Weekly check-in	15 minutes	What worked, what did not, next week's lessons
Pilot week 4	Mid-pilot workshop	60 minutes	Teaching conditionals and loops; managing different paces; project start-up (Classes 6-8)
Pilot week 8	Assessment workshop	60 minutes	Using the rubrics and journal; preparing the term summary
Full year	Termly workshop	90 minutes per term	New units, AI chapters, project kits, sharing practice across teachers
Ongoing	Question channel and short videos	As needed	Quick help and how-to clips for each activity

Teachers who complete the orientation and the termly workshops receive a CodeSchool certificate of completion.

Orientation agenda (2 hours): welcome and goals (10 minutes); CBSE CT&AI in one page (15); the lesson routine, experienced as students (35); planning one lesson from the script (20); the journal and rubrics (20); timetable and support (10); questions (10).

Contact: Anoop Dobhal, CodeSchool · Phone / WhatsApp: [Phone] · Email: [Email]